

Poisson Process

Dr. Collins Odhiambo

Institute of Mathematical sciences, Strathmore University

codhiambo@strathmore.edu

July 1, 2022

- 1 Overview of Poisson Process
 - Counting process
- 2 Poisson Process
- 3 Deriving Poisson Process
 - Chapman-Kolmogorov Postulates
 - Derivation of Poisson Process

Overview of Poisson Process

A counting process is a random process $N(t)$, $t \geq 0$, such that

- $N(t)$ is a nonnegative integer for each t ;
- $N(t)$ is nondecreasing in t ; and
- $N(t)$ is right-continuous.

Counting Process

The third condition is merely a convention: if the first two events happens at $t = 2$ and $t = 3$ we want to say $N(2) = 1$, $N(3) = 2$, $N(t) = 1$ for $t \in (2, 3)$, and $N(t) = 0$ for $t < 2$. Thus $N(t) - N(s)$ represents the number of events in $(s, t]$.

Counting processes deal with the number of distinct occurrences of events over time. An example of a counting process is the number of job arrivals to a queue over time. If a process has the Markov property, it is said to be a Markov counting process.

Poisson Process

A Poisson process with rate (or intensity) $\lambda > 0$ is a counting process $N(t)$ such that

- $N(0) = 0$;
- It has independent increments: if $(s_1, t_1] \cap (s_2, t_2] = \emptyset$, then $N(t_1) - N(s_1)$ and $N(t_2) - N(s_2)$ are independent; and
- Number of events in any interval of length t is $\text{Poisson}(\lambda_t)$.

Poisson Process

Poisson processes are used to model continuous-time arrivals into a system (customers, orders, signals, packets), and can be thought of as continuous-time analogs of the Bernoulli processes. In a Bernoulli process the arrival times are fixed and discrete, while for general arrival processes the arrival times themselves are random variables, and form a random process.

Arrival Process

An arrival process is a sequence of increasing rv's $0 < S_1 < S_2 < \dots$, where the inequality $S_j < S_{j+1}$ is meant in the sense that the difference $S_{j+1} - S_j$ is a positive rv X (i.e. $F_X(0) = 0$). The rv's $S_j, j = 1, 2, \dots$ are called arrival epochs, and they give the times of first, second, etc. arrivals.

- One can use an arrival process to model any random repeating phenomenon, where the epochs S_j will represent the times when the phenomenon occurs.
- For a continuous arrival process the arrival epochs are continuous rv's, and hence the probability of an arrival at any fixed time is zero.
- To fully describe the arrival process one needs to specify the joint distribution of $S_1, S_2, \dots$.

Chapman-Kolmogorov Postulates

Consider a positive real line and a collection of random points on this line. Let $N(t, t + h)$ denote number points in an interval of length h .

- $N(0, 0) = 0$, that is initially no point has occurred.
- Number of occurrences of points in two non-overlapping intervals are independent.
- Probability that occurrence of one point in a small interval of duration of length h is proportional to h . $Pr(N(t, t + h) = 1) = \lambda h + 0(h)$.
- The probability for more than one occurrence of points in a small interval of duration of length h is zero, that is,
 $Pr(N(t, t + h) > 1) = 0(h)$.
- Thus $Pr(N(t, t + h) = 0) = 1 - \lambda h + 0(h)$.

where λ is a constant of proportionality and $\lim_{h \rightarrow 0} \frac{0(h)}{h} \rightarrow 0$. Using the first 3 postulates, we can derive $Pr(\text{number of points in an interval } (0, t) = n)$

$$= P_t(n) = \frac{e^{-\lambda h} (\lambda h)^n}{n!}$$

Derivation of Poisson Process

Define i.e. $P_n(t)$ is the number of n counts at time t , then

$$\begin{aligned}P_n(t+h) &= P_n(t)Pr\{N(t, t+h) = 0\} + P_{n-1}(t)Pr\{N(t, t+h) = 1\} \\&\quad + P_{n-2}(t)Pr\{N(t, t+h) = 2\} + \dots \\&= P_n(t)(1 - \lambda h + 0(h)) + P_{n-1}(t)(\lambda h + 0(h)) + 0(h) \\&= P_n(t) - P_n(t)\lambda h + P_n(t)0(h) + P_{n-1}(t)\lambda h + P_{n-1}(t)0(h) + 0(h)\end{aligned}$$

Reorganizing the expression yields

$$P_n(t+h) - P_n(t) = -P_n(t)\lambda h + P_n(t)0(h) + P_{n-1}(t)\lambda h + P_{n-1}(t)0(h) + 0(h)$$

Dividing across with h and introducing limits

Derivation of Poisson Process

$$\lim_{h \rightarrow 0} \frac{P_n(t+h) - P_n(t)}{h} = \frac{d}{dt} p_n(t)$$
$$\lim_{h \rightarrow 0} \frac{-P_n(t)\lambda h + P_n(t)0(h) + P_{n-1}(t)\lambda h + P_{n-1}(t)0(h)}{h}$$
$$\frac{d}{dt} p_n(t) = P'_n(t) = -P_n(t)\lambda + P_{n-1}(t)\lambda$$

Solving first order ordinary differential equation

$$P'_n(t) = -P_n(t)\lambda + P_{n-1}(t)\lambda$$

$$P'_n(t) + P_n(t)\lambda = P_{n-1}(t)\lambda$$

Solving the differential equation by introducing $e^{\lambda t}$

$$\frac{d}{dt} e^{\lambda t} P_n(t) = P_{n-1}(t)\lambda$$

Derivation of Poisson Process

Reorganizing the expression

$$\frac{d}{dt}e^{\lambda t}P_n(t) = \lambda P_{n-1}(t)$$
$$\int de^{\lambda t}P_n(t) = \int \lambda P_{n-1}(t)dt$$

Solving the integration

$$e^{\lambda t}P_n(t) = \lambda t P_{n-1}(t)$$

Solving the difference equation iteratively. Let $n = 1$

$$e^{\lambda t}P_1(t) = \lambda t P_0(t) = \frac{\lambda t P_0(t)}{1!}$$

For $n = 2$, we have

$$e^{\lambda t}P_2(t) = \lambda t P_1(t) = \frac{(\lambda t)^2 P_0(t)}{2!}$$

Derivation of Poisson Process

For $n = 3$, we have

$$e^{\lambda t} P_3(t) = \lambda t P_2(t) = \frac{(\lambda t)^3 P_0(t)}{3!}$$

Therefore, generalizing for n , we have

$$e^{\lambda t} P_n(t) = \frac{(\lambda t)^n P_0(t)}{n!}$$

But the sum of probabilities $P_0(t) = 1$ and rearranging the expression, we have

$$P_n(t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

The End